

Biostatistics 1 - Assignment 2

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Preparing for analysis and cleaning the data

A new dataframe was created containing only the variables relevant for the analysis. The variables included are:

- id
- age0
- cholmmol
- sbp10
- bmi
- smokerf
- arcus0
- dibpat0f
- agegroup

After that all complete cases were extracted from the cleaned dataset.

1 Task 1

	Overall
n	3139
id (mean (SD))	10480.53 (5882.89)
agegroup (%)	
[39,45)	1440 (45.9)
[45,55)	1380 (44.0)
[55,60]	319 (10.2)
age0 (mean (SD))	46.28 (5.52)
cholmmol (mean (SD))	5.80 (1.10)
sbp10 (mean (SD))	12.86 (1.51)
bmi (mean (SD))	24.52 (2.56)
smokerf = Yes (%)	1494 (47.6)
arcus0 (mean (SD))	0.30 (0.46)
dibpat0f = A (%)	1582 (50.4)
chd69 (mean (SD))	0.08 (0.27)

2 Taks 2

2.1 What is the outcome we are interested in?

We are interested in the outcome coronary heart disease

2.2 What are the known risk factors for our outcome of interest?

The known factors that cause risk for the particular disease are separated into two groups (modifiable and non-modifiable) (Centers for Disease Control and Prevention 2024).

Modifiable risk factors are:

- smoking
- physical inactivity
- unhealthy diet
- high blood pressure
- unhealthy blood cholesterol levels

Non-modifiable risk factors are:

- sex
- ethnicity
- age
- family history (genetics)

2.3 How many people are included?

The data that is used for this analysis have 3139 observations.

2.4 What is the overall prevalence of the disease in our cohort?

To find the overall prevalence or the percentage of people who have the disease, the mean of the binary outcome variable is calculated.

```
mean(dc$chd69)
```

```
[1] 0.08091749
```

Meaning that the overall prevalence is 8%.

3 Task 3

3.1 Part 1

Before building a predictive model, a correlation matrix was calculated to check for variables with strong correlation. The threshold for strong correlation is 0.8.

Table 1: Correlation Matrix of Numeric Variables

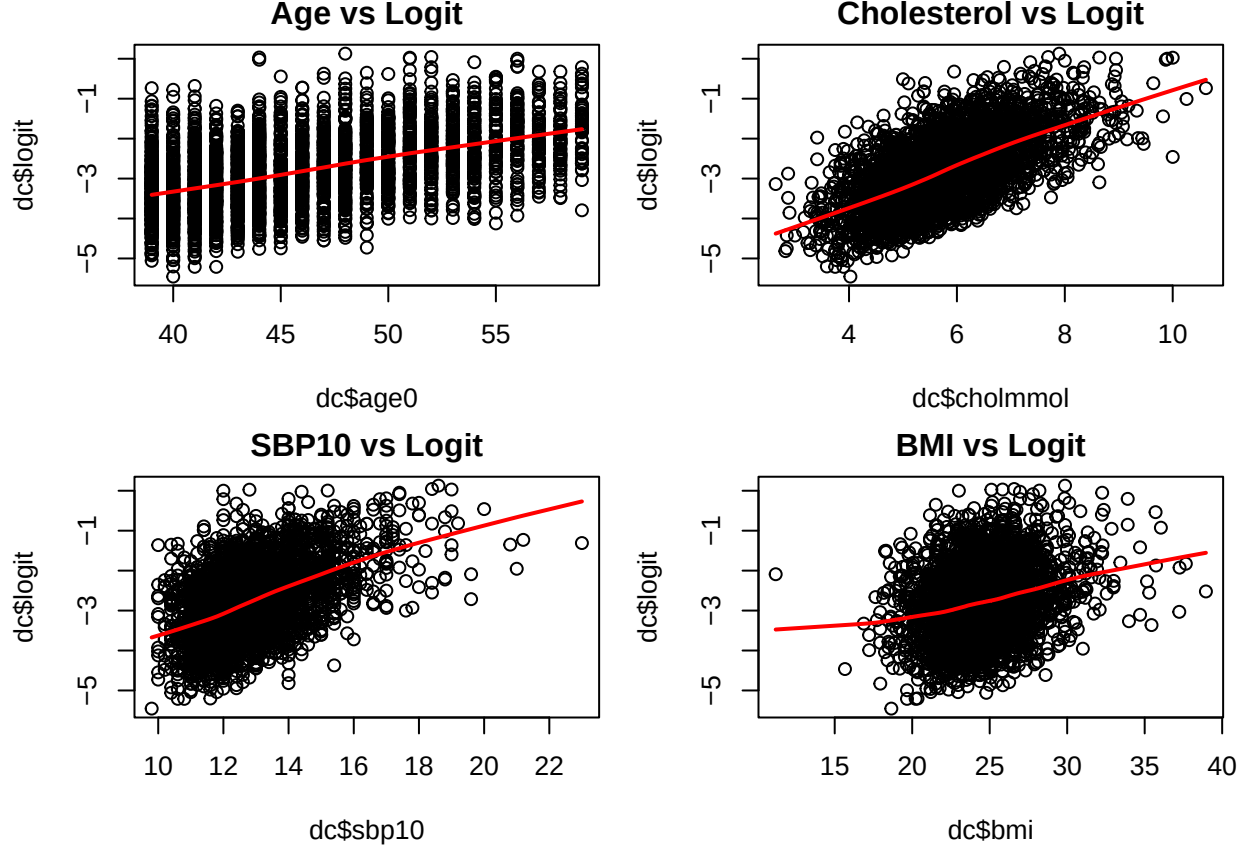
	id	age0	cholmmol	sbp10	bmi	arcus0	chd69
id	1.00	-0.05	0.06	-0.05	0.03	-0.02	0.00
age0	-0.05	1.00	0.09	0.17	0.03	0.19	0.12
cholmmol	0.06	0.09	1.00	0.12	0.07	0.13	0.15
sbp10	-0.05	0.17	0.12	1.00	0.29	0.03	0.13
bmi	0.03	0.03	0.07	0.29	1.00	-0.03	0.06
arcus0	-0.02	0.19	0.13	0.03	-0.03	1.00	0.07
chd69	0.00	0.12	0.15	0.13	0.06	0.07	1.00

The matrix shows that none of the variables exceed the threshold of 0.8, therefore there is no risk for multicollinearity in this dataset. Meaning that it is not necessary to remove any of the variables at this stage of the analysis, since every variable contributes uniquely.

A logistic regression model was selected for this analysis because the outcome variable is binary and diagnostic checks indicated no evidence of multicollinearity among the predictors.

The next step was to fit the full logistic regression model, which included all available predictors in the cleaned dataset. This model was then compared to a reduced model obtained using backward stepwise selection, where predictors were removed based on their contribution to model fit. In our case the function only excluded one of the initial variables (agegroup). The result yields that the step model had better performance compared to the original one. This is concluded since AIC value is lower than in the full model. This result is also conceptually reasonable since agegroup and age0 represent the same underlying information and including both in the model introduces redundancy. Moreover, age0 is more informative as a continuous variable and agegroup was appropriately removed by the stepwise algorithm.

To assess whether the continuous predictors satisfy the logistic regression assumption of linearity in the logit, scatterplots of each predictor against the predicted log-odds were created. They show that the requirement for linearity is met and step-wise logistic regression could be used.



Below the coefficients of the step-wise model could be seen. Only one of them is not statistically significant.

Table 2: Logistic Regression Coefficient Table

Variable	Estimate	Std_Error	Z_value	P_value
(Intercept)	-12.2494	0.9865	-12.417	< 2e-16
age0	0.0573	0.0123	4.669	3.03e-06
cholmmol	0.4014	0.0600	6.686	2.30e-11
sbp10	0.1820	0.0416	4.380	1.18e-05
bmi	0.0593	0.0266	2.229	0.0258
smokerfYes	0.6002	0.1419	4.230	2.34e-05
arcus0	0.2204	0.1434	1.537	0.1243
dibpat0fA	0.7014	0.1451	4.834	1.34e-06

Therefore, the chosen step-wise model has the following form:

$$\begin{aligned} \text{logit}(P(\text{CHD}_i = 1)) = & \beta_0 + \beta_1 \cdot \text{age0}_i + \beta_2 \cdot \text{cholmmol}_i + \beta_3 \cdot \text{sbp10}_i \\ & + \beta_4 \cdot \text{bmi}_i + \beta_5 \cdot \text{smoker}_i + \beta_6 \cdot \text{arcus0}_i + \beta_7 \cdot \text{dibpat0fA}_i. \end{aligned}$$

Furthermore, we compared the AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) values of the full model and the stepwise model. Both criteria favored the stepwise model, as it achieved lower AIC and BIC values than the full model. This indicates that the stepwise model provides a better balance between goodness of fit and model complexity.

Table: Comparison of AIC and BIC Between Models

Metric	Value
AIC (Full Model)	1597.072
AIC (Stepwise Model)	1594.227
BIC (Full Model)	1657.589
BIC (Stepwise Model)	1642.640

3.2 Part 2

Several interaction terms were tested to assess whether the effect of one predictor on CHD depended on another predictor. The interactions were selected based on clinical reasoning and included age with cholesterol, age with systolic blood pressure, smoking with cholesterol, smoking with systolic blood pressure, and BMI with cholesterol.

First, a model containing all interaction terms was fitted. Then, individual models were fitted in which each interaction term was added separately together with all main predictors. For each model, AIC values were compared with those from the stepwise model and the significance of the interaction term was evaluated.

Across these models, most interaction terms did not improve model performance and were not statistically significant. Only the BMI with cholesterol interaction improved the model by lowering the AIC and showing statistical significance. This indicates that, while most predictors act independently in this dataset, the BMI with cholesterol interaction slightly modify the effect of one predictor on CHD.

Table 4: Table: Comparison of AIC Between Interaction Model and Stepwise Model

Model	AIC
Interaction Model ($\text{bmi} \times \text{cholmmol}$)	1588.306
Stepwise Model	1594.227

Table 5: Logistic Regression Coefficients for the Model Including the BMI $\times$ Cholesterol Interaction

Variable	Estimate	Std_Error	Z_value	P_value
(Intercept)	-21.2833	3.3053	-6.439	1.20e-10
age0	0.0564	0.0123	4.595	4.32e-06
cholmmol	1.8755	0.5187	3.616	0.00030
sbp10	0.1868	0.0416	4.492	7.05e-06
bmi	0.4195	0.1277	3.284	0.00102
smokerfYes	0.5978	0.1420	4.211	2.55e-05
arcus0	0.2190	0.1438	1.523	0.12771
dibpat0fA	0.7266	0.1456	4.989	6.07e-07
cholmmol:bmi	-0.0591	0.0206	-2.864	0.00418

In the Table 5 can be seen that the interaction is statistically significant.

Furthermore, a likelihood ratio test was conducted to compare the stepwise-selected main-effects model with the model that included the BMI $\times$ cholesterol interaction term.

Table 6: Likelihood Ratio Test Comparing Models With and Without the BMI $\times$ Cholesterol Interaction

Model	Residual_Df	Residual_Deviance	Df	Deviance	P_value
Stepwise Model (no interaction)	3131	1578.2			
Interaction Model (with bmi:cholmmol)	3130	1570.3	1	7.9208	0.004887

This result suggests that the effect of cholesterol on CHD risk depends on BMI, and that including the bmi $\times$ cholmmol interaction provides additional information.

After testing if the interaction model is suitable for this analysis the model that is chosen is:

$$\begin{aligned} \text{logit}(P(\text{CHD}_i = 1)) = & \beta_0 + \beta_1 \cdot \text{age0}_i + \beta_2 \cdot \text{cholmmol}_i + \beta_3 \cdot \text{sbp10}_i \\ & + \beta_4 \cdot \text{bmi}_i + \beta_5 \cdot \text{smoker}_i + \beta_6 \cdot \text{arcus0}_i \\ & + \beta_7 \cdot \text{dibpat0fA}_i + \beta_8 \cdot (\text{cholmmol}_i \cdot \text{bmi}_i). \end{aligned}$$

3.3 Part 3

```
dc$predicted_risk <- predict(interaction_model_bmi_cholmmol, newdata = dc, type = "response")
kable(head(dc), digits=3)
```

id	agegroup	age0	cholmmol	sbp10	bmi	smokerf	arcus0	dibpat0f	chd69	logit	predicted_risk
2001	[45,55)	49	5.769	11.0	19.790	Yes	0	A	0	- 2.651	0.060
2002	[39,45)	42	4.538	15.4	22.957	Yes	1	A	0	- 2.337	0.076
2003	[39,45)	42	4.641	11.0	23.628	No	0	B	0	- 4.579	0.009
2004	[39,45)	41	3.385	12.4	23.111	Yes	0	B	0	- 4.316	0.010
2005	[55,60]	59	6.538	14.4	21.523	Yes	1	B	1	- 1.529	0.187
2006	[39,45)	44	4.667	15.0	27.667	No	0	B	0	- 3.486	0.037

References

Centers for Disease Control and Prevention. 2024. "Heart Disease: Risk Factors." <https://www.cdc.gov/heart-disease/risk-factors/index.html>.